Panfeng Jiang, Undergraduate

jiangpf2022@shanghaitech.edu.cn
jiangpf2022@berkeley.edu

ShanghaiTech University
School of Information Science and Technology
Computer Science
Recent Works:Event-based vision, SLAM, Computer Vision
GPA:3.77/4.0 , **Ranking 8%-15%** in SIST College in the comprehensive evaluation

Research Publications

1. S. Hang, L. Tao, **Panfeng. Jiang**, G. Ling, and K. Laurent, "Led matrix-based simultaneous localization and data transmission with an event camera," *(Still Modifying)*, 2023.

- Develop the detection part and new modules for event-based localization, spatiotemporal tracking, and real-time decoding.
- Participated in the process of encoding-decoding, tracking algorithm designing and experimental testing.
- Our efforts highlighted the practical advantages of our solution by using event-based camera, particularly in achieving 64 Kbps of bandwidth and centimeter-level positioning accuracy, beyond the capabilities of CMOS cameras.

Research Experiences

2023.4 – now **Mobile Perception Lab, ShanghaiTech University, China**

Working Experiences and Social Participation

2023.7 – 2023.8 Served as a **teaching assistant** in the New Oriental TOEFL course, responsible for course quality supervision and feedback, after-class homework assignments and corrections, and course answering for classmates

2023.8-11 Served as a **volunteer** in the summer camp of ShanghaiTech University , Shanghai Planetarium , Shanghai Library and the 27.5km mark of the 2023 Shanghai Marathon

2023.12 Served as a **referee** in Changsha Rubik's Cube Open 2023

Ranking of the Comprehensive Evaluation

2022-2023 **Ranking 8%-15% in SIST College**

Skills

Languages Familiar with English, Mandarin Chinese(mother language).

Coding Python,C/C++, …

Experimental skills Usage of ROS (Robot Operating System) , event-based camera, motion capture system.

Knowledge Algorithms and Data Structure, Machine Learning, Visual SLAM, Event-based Vision, Simple Geometry Algebra(only a little),Deep Learning (only a little).

Miscellaneous Experience

Awards and Achievements

2023.12 **Department Prize for Outstanding Student Performance**, ShanghaiTech University.

Miscellaneous Experience (continued)

Certification

2022.12 **CET-4 642**

2023.3 **Certified as RHCSA (Red Hat® Certified System Administrator)**. Awarded by Red Hat,Inc.

Certified as RHCE (Red Hat® Certified Engineer). Awarded by Red Hat,Inc.

2023.9 **TOFEL iBT 91**

2023.12 **CET-6 573**

Relevant Course Projects

Computer Visioin

- **Disentangling Object Motion for Self-supervised Depth Estimation**(Self-supervised depth estimation heavily suffers from the mismatch of the photometric consistency caused by moving objects. And we trying to mitigate the negative impact caused by this.)

SLAM

- **Instance-level Object Pose Estimation with an Event Camera**(Event camera is able to track fast object motion, and based on this advantage, I am going to track the 6DoF object motion given the object CAD model by using convex-hull detected and ICP algorithm based methods.)
- Toy realization of Ordinary& Scale-aware & Jump-aware Pose Graph Optimization,Visual Odometry, Particle Filter and Five-point Minimal Solver for Event Camera Relative Motion Estimation.

Machine Learning

- **Use multiple models for quality prediction** (discuss the advantages and disadvantages of logistic regression, random forest, XGBoost models, and combine principal component analysis and Huber Loss for model performance optimization)

Grades of Some Completed Courses

Overall GPA

- 3.77/4.0(Ranking in all Computer Science students:16/169,Ranking in all SIST students:27/243).

2022 Fall

- **Introduction to Information Science and Technology**(A-), Introduction to Chinese Civilization(B+), United States 1770-1920:from the Revolution to the Great Migration(A-),**Mathematical Analysis I**(A),**Linear Algebra I**(A-),Military Course in Theory, Fitness and Health Military Course in Practice(A+)

2023 Spring

- General Chemistry I Lab(B+), General Chemistry I(B+), Introduction to Programming(A+), **Discrete Mathematics**(A), Introduction to Economics: Information Technology(A-), Design Thinking: Innovative Solutions in Entrepreneurship and Management(B+), Morality and the Rule of Law(B+), **Mathematical Analysis II**(A) SWIMMING(A)

2023 Fall

- **Algorithm and Data Structure**(A+),**Computer Vision I**(A),Signals and Systems(B+),Signals and Systems lab(A+),Introduction to Statistics and Probability(B+),General Physics(A),1001 inventions: scientific discovery and enlightenment of Islam at the era of Enlightenment(A+),Table Tennis(B+).

2024 Spring

- **Computer Architecture I**(A),**Computer Architecture I Lab**(A),**Simultaneous Localization and Mapping**(A),**Introduction to Machine Learning**(A+),Table Tennis(A-),Classical Music Appreciation(A-),Literature and Film:Hong Kong and Taiwan(A),Elementary Principles of Marxism(A+)